

Tangents to parametric curves — 7.2

If position is given by $(x(t), y(t))$,

then velocity is $(x'(t), y'(t))$

To find the slope of the curve at a certain point take change in output over change in input, then take a limit

$$\text{slope} = \lim_{h \rightarrow 0} \frac{y(t+h) - y(t)}{x(t+h) - x(t)} = \lim_{h \rightarrow 0} \frac{\frac{y(t+h) - y(t)}{h}}{\frac{x(t+h) - x(t)}{h}} = \frac{y'(t)}{x'(t)}$$

Example: Find a line tangent to the unit circle at $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$.

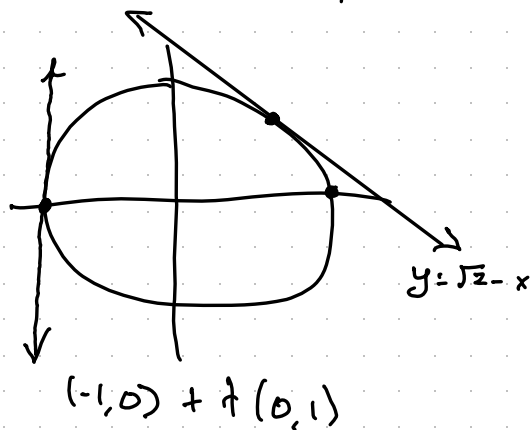
We can parameterize the unit circle as $(\cos(t), \sin(t))$, where $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$ is the output for $t = \frac{\pi}{4}$.

Then slope is $\frac{y'(t)}{x'(t)} = \frac{\cos(t)}{-\sin(t)}$, at $t = \frac{\pi}{4}$,

this is $\frac{\frac{\sqrt{2}}{2}}{-\frac{\sqrt{2}}{2}} = -1$.

Tangent line is then

$$y - \frac{\sqrt{2}}{2} = -1(x - \frac{\sqrt{2}}{2}) \iff y = \sqrt{2} - x$$



Parametrically, the tangent line to

$r(t) = (x(t), y(t))$ at $t=a$ is given by

$$r(a) + r'(a)(t-a) = (x(a) + x'(a)(t-a), y(a) + y'(a)(t-a))$$

For the previous example, $r(t) = (\cos(t), \sin(t))$

so that $r'(t) = (-\sin(t), \cos(t))$, which gives

$$r\left(\frac{\pi}{4}\right) + r'\left(\frac{\pi}{4}\right)(t - \frac{\pi}{4}) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) + \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)(t - \frac{\pi}{4})$$

$$= \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}(t - \frac{\pi}{4}), \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}(t - \frac{\pi}{4})\right)$$

Equally good to say the tangent line is

$$r(a) + r'(a)t, \quad \text{in our case } \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) + t\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right).$$